1 Introduction and main results

1.1 The SYK model

The Sachdev–Ye–Kitaev (SYK) model is a quantum-mechanical model of N fermions with random interactions between all pairs of particles. It originates in the work of Sachdev and Ye on random quantum magnets [**?**] and in Kitaev's subsequent formulation in terms of Majorana fermions [**?**]. Later work by Maldacena, Stanford, and others clarified its large-N behavior and revealed deep connections to quantum chaos and two-dimensional quantum gravity [**?**]. Although the model has strong interactions, it can still be tractable mathematically, which makes it a useful model for studying quantum many-body systems, disorder, quantum chaos, and quantum gravity.

The behavior of the SYK model depends strongly on how the inverse temperature $\beta = \beta_N$ scales with N. For fixed β, one obtains the standard finite-temperature large-N regime; when $1 \ll \beta \ll N$, the system enters the conformal regime, characterized by strong interactions and non-Fermi-liquid behavior. At lower temperatures, $N \ll \beta$, the quantum-Schwarzian regime emerges, where finite-N effects and quantum fluctuations of the Schwarzian mode become essential. Finally, when β is exponentially large in N, the partition function is mainly determined by the eigenvalues near the edge of the spectrum. In this regime, individual energy levels become important, and random matrix theory (RMT) describes their fluctuations.

Previous mathematical work has extensively studied the limiting free energy of the SYK model [**?**, **?**]. Moreover, Anschuetz, Gamarnik, and Kiani [**?**] established a universality result showing that the limiting free energy is the same for a broad class of interaction distributions. This naturally leads us to ask how physical quantities change as the interaction distribution is varied. Motivated by this question, we study fluctuations of the free energy and show that, for sufficiently small β, the suitably rescaled free energy converges to a Gaussian distribution.

Fix an even integer $q \geq 4$, and let N be even. Write

$$\mathcal{I}_N := \{I \subset [N] : |I| = q\}, \qquad \sigma_N^2 := \frac{(q-1)!}{N^{q-1}},$$

and let $G := (g_I)_{I \in \mathcal{I}_N}$ be independent standard Gaussian random variables. Let $\psi_1, \ldots, \psi_N$ be Majorana matrices satisfying $\{\psi_i, \psi_j\} = 2\delta_{ij}\mathbb{I}$. For $I = \{i_1 < \cdots < i_q\}$ set

$$\psi_I := \mathrm{i}^{q/2} \psi_{i_1} \cdots \psi_{i_q}.$$

Then ψ_I is Hermitian, $\psi_I^2 = \mathbb{I}$, and $\|\psi_I\| = 1$. Define

$$H_N(G) := \sigma_N \sum_{I \in \mathcal{I}_N} g_I \psi_I, \qquad Z_N(z; G) := \operatorname{tr} e^{-zH_N(G)}.$$

Here $\operatorname{tr} := 2^{-N/2} \operatorname{Tr}$ denotes the normalized trace; in particular, $Z_N(0; G) = 1$. This normalization does not affect the relative fluctuations studied below. We sometime write $Z_N(\beta) := Z_N(\beta; G)$ for brevity. For real β, set

$$Y_N(\beta) := Y_N(\beta; G) := \frac{Z_N(\beta; G)}{\mathbb{E} Z_N(\beta)}.$$

Thus $\mathbb{E} Y_N(\beta) = 1$. Throughout, all limits are taken as $N \to \infty$ through even integers.

1.2 Main results

It is studied the limiting behavior of the logarithmic partition function in high-temperature regime:

Theorem 1.1. *There exists $\tilde{\beta}_q > 0$ such that for any $\beta \in [-\tilde{\beta}_q, \tilde{\beta}_q]$, there exists $\ell_q(\beta)$ such that*

$$\lim_{N \to \infty} N^{-1} \log Z_N = \lim_{N \to \infty} N^{-1} \log \mathbb{E}[Z_N] = \ell_p(\beta) \qquad \text{a.s.}$$

Moreover, $\ell_p(\beta)$ is differentiable over $(-\tilde{\beta}_q, \tilde{\beta}_q)$.

Our main result is the following fluctuation resutls:

Theorem 1.2. *There exists $\beta_q \in (0, \tilde{\beta}_q]$ such that for any $\beta \in (-\beta_q, \beta_q)$,*

$$N^{q/2-1}\left(Y_N(\beta) - 1\right) \Longrightarrow \mathcal{N}\left(0, \frac{q!}{2}\beta^2(\ell_q'(\beta))^2\right). \tag{1}$$

Moreover,

$$\frac{Z_N(\beta) - \mathbb{E}Z_N(\beta)}{\sqrt{\operatorname{Var} Z_N(\beta)}} \Longrightarrow \mathcal{N}(0,1). \tag{2}$$

Corollary 1.3. *With the same β_p as in Theorem 1.2, for any $\beta \in (-\beta_q, \beta_q)$,*

$$N^{q/2-1}\left(\log Z_N(\beta) - \mathbb{E}\log Z_N(\beta)\right) \Longrightarrow \mathcal{N}\left(0, \frac{q!}{2}\,\beta^2(\ell_q'(\beta))^2\right).$$

1.3 Previous work

Rigorous results on the high-temperature annealed and quenched free-energy limits were recently obtained in [**?**]. More recently, He [**?**] identified the fixed-temperature free-energy limit of the Gaussian SYK model with the Schwinger–Dyson pressure and used this result to determine the spectral edge. These works focus on the leading-order behavior of the free energy, while the purpose of the present paper is to study its leading fluctuations in the high-temperature regime.

Feng, Tian, and Wei [**?**] proved a central limit theorem for linear eigenvalue statistics of the Gaussian SYK Hamiltonian. Their result applies to a fixed test function f, under suitable regularity assumptions on f and f'. In their normalization, however, the partition function corresponds to

$$f_{N,\beta}(x) := e^{-\beta\tau_N x}, \qquad \tau_N \asymp \sqrt{N},$$

so both the test function and its derivative depend on N. Therefore, their fixed-test-function theorem does not apply directly to $Z_N(\beta)$.

1.4 Outline of the proof

Recall that $Y_n(\beta) := Z_N(\beta)/\mathbb{E}Z_N(\beta)$. By applying the chaos expansion, since Clifford sign-gauge symmetry eliminates the first Gaussian chaos and identifies the second chaos exactly, we have the following:

$$Y_N(\beta) - 1 = A_N(\beta)Q_N + \mathcal{R}_N(\beta),$$

where

$$A_N(\beta) := \frac{\beta}{2}\partial_\beta \log \mathbb{E}Z_N(\beta), \qquad Q_N := \frac{1}{|\mathcal{I}_N|}\sum_I (g_I^2 - 1). \tag{3}$$

Here $\mathcal{R}_N$ is the sum of the Gaussian chaoses of Y_N of orders at least three. Thus the leading term is a normalized sum of $\mathcal{I}_N$ independent centered variables. The main task is to prove

$$\|\mathcal{R}_N(\beta)\|_2^2 = o(N^{-(q-2)}).$$

Indeed, a Duhamel derivative bound and the sign-gauge parity rule give

$$\sum_{r\geq 3}\|P_r Y_N(\beta)\|_2^2 \leq \sum_{r\geq 3}(\beta\sigma_N)^{2r} \sum_{\substack{|\alpha|=r \\ \alpha \text{ even}}} \frac{1}{\alpha!}.$$

Using a combinatorial argument, we have that the entire higher-order sum is bounded by $O(N^{-3(q/2-1)}) = o(N^{-(q-2)})$.

Moreover, the annealed free-energy input and convexity imply

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2}\ell_q'(\beta).$$

Since $M_N \sim N^q/q!$, the classical central limit theorem for Q_N gives the first claim.

To obtain the second claim, the logarithmic fluctuation, observe that

$$\begin{aligned}
&N^{q/2-1}\big(\log Z_N(\beta) - \mathbb{E}\log Z_N(\beta)\big) \\
&= N^{q/2-1}\big(\log Y_N(\beta) - (Y_N(\beta)-1)\big) + N^{q/2-1}(Y_N(\beta)-1) - N^{q/2-1}\mathbb{E}\log Y_N(\beta).
\end{aligned}$$

The Taylor expansion $\log(1+x) - x = O(x^2)$, together with the first claim, shows that the first term on the right-hand side converges to zero in probability. The Gaussian concentration estimates (**??**) and (**??**) control the tails of $\log Y_N(\beta)$. Moreover, these estimats give

$$\mathbb{E}\log Y_N(\beta) = O\Big(N^{2-q}\sqrt{\log N}\Big) = o\Big(N^{1-q/2}\Big).$$

Therefore, we conclude the second claim.